YASH DHINGRA

Lead Software Engineer • C++ / Python • Systems & Media Backend

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CORE COMPETENCIES

- Systems Architecture
- Low-Latency Engineering
- Media Pipeline QC
- Distributed Systems
- AI Agents & Orchestration
- Technical Leadership

TECHNICAL SKILLS

Languages & Core
C++ (11/14/17/20), Python, C, Go, Java, Bash, SQL

Systems Engineering
Linux internals, pthreads, IPC, sockets, lock-free structures

Video & Media Stack
HLS, DASH, FFmpeg, GStreamer, DRM, secure pipelines

Backend & Messaging
Flask, FastAPI, MQTT (Mosquitto), REST

Profiling & Debug
GDB, Valgrind, perf, AddressSanitizer, strace

Architecture & Build
CMake, Make, STL, Boost, distributed systems

Cloud & DevOps
AWS (EC2, Lambda, S3, ECR), Docker, CI/CD

AI & Workflows
CrewAI, LangGraph, LangChain, multi-agent orchestration, generative & diffusion models

EDUCATION

B.Tech, Information Technology
YMCA Univ. of Science & Technology
2017–2021 | Faridabad, HR

CERTIFICATIONS

SWC Professional
Top-tier Samsung Software Competency credential

LANGUAGES

English (Professional)
Hindi (Native)

PROFESSIONAL SUMMARY

Lead Software Engineer and hands-on C++ & Python engineer with 5 years at Samsung building high-performance media backends. Automated HLS stream monitoring across Samsung TV Plus' 4,000+ live channels and took the media QC automation platform from proof-of-concept to production. Lead an 8-engineer team running it at 24/7 uptime for overseas operations, while driving multi-agent AI orchestration for content analysis.

PROFESSIONAL EXPERIENCE

Lead Software Engineer — May 2022 – Present
Samsung • Noida, India

Content Engineering Team — Media QC & Monitoring (Jul 2023 – Present)

- **QC Automation:** Took the QC automation framework from proof-of-concept to a production platform — Python, Flask, and C++ — cutting manual operational effort across the media stack by ~80%.
- **HLS Monitoring Pipeline:** Built and fully automated an HLS stream-monitoring tool for the operations team across Samsung TV Plus' 4,000+ live channels — analyzing streams and raising real-time alarms on faults, on Mosquitto (MQTT), Flask/FastAPI, and AWS.
- **Real-Time Monitoring & Reliability:** Engineered continuous 24/7 live-stream monitoring and sustain platform uptime for overseas operations — including maintenance of physical servers in a remote lab / server farm — with the C++ stack tuned (Valgrind, GDB, perf) for ~20% lower latency under 1,000+ RPS.
- **Team Leadership:** Lead an 8-engineer team across periodic releases, QA fixes, on-time delivery, and ongoing development of new test cases.
- **Multi-Agent AI:** Built a multi-agent system for automated content analysis, integrating LLM and generative components into the media-QC workflow.

Application Software Team — SES Cloud (May 2022 – Jun 2023)

- **SES Cloud Backend:** Core backend developer on the Samsung SES cloud service — built new feature requirements end to end, owned unit testing, defect fixing, and on-time release delivery.

Software Engineer — Jul 2021 – Apr 2022
Tata Consultancy Services (TCS) • India

- Developed and maintained backend service modules and REST APIs within an Agile delivery team; wrote unit/integration tests and resolved defects through structured debugging — the systems-engineering foundation carried into media-backend work.

KEY PROJECTS

AI Multi-Agent Orchestration System

- Designed and built a multi-agent orchestration system using CrewAI, LangGraph, and LangChain — coordinating LLM-driven agents with tool-use and stateful workflows for automated content-analysis and QC tasks.

Generative Media Automation Pipeline

- Prototyped media-automation workflows using generative diffusion models to explore next-generation automated-testing layers for smart-ecosystem infrastructure.

AWARDS & ACHIEVEMENTS

- **Best Growth Project Award (2024, 2025):** for architecting Samsung's high-concurrency media-automation pipeline.
- **NPCI Hackathon Winner:** for high-throughput, low-latency distributed algorithms.